

$$(iii) A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$$

(A)

$$[A|I] = \left[\begin{array}{ccc|ccc} -1 & 2 & -3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & -2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\underline{v_1' = (-1)v_1} \rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & -1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 4 & -2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{array}{l} \underline{v_2' = -2v_1 + v_2} \\ \underline{v_3' = -4v_1 + v_3} \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & -1 & 0 & 0 \\ 0 & 5 & -6 & 2 & 1 & 0 \\ 0 & 6 & -7 & 4 & 0 & 1 \end{array} \right]$$

$$\underline{v_2' = -v_2 + v_3} \rightarrow \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & -1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 \\ 0 & 6 & -7 & 4 & 0 & 1 \end{array} \right]$$

$$\begin{array}{l} \underline{v_1' = 2v_2 + v_1} \\ \underline{v_3' = -6v_2 + v_3} \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 3 & -2 & 2 \\ 0 & 1 & -1 & 2 & -1 & 1 \\ 0 & 0 & -1 & -8 & 6 & -5 \end{array} \right]$$

$$\underline{v_3' = (-1)v_3} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 3 & -2 & 2 \\ 0 & 1 & -1 & 2 & -1 & 1 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right]$$

$$\begin{array}{l} \underline{v_1' = -v_3 + v_1} \\ \underline{v_2' = v_2 + v_3} \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -5 & 4 & -3 \\ 0 & 1 & 0 & 10 & -7 & 6 \\ 0 & 0 & 1 & 8 & -6 & 5 \end{array} \right]$$

$$\begin{array}{l} \underline{v_1' = -v_3 + v_1} \\ \underline{v_2' = v_2 + v_3} \end{array} \rightarrow \begin{array}{l} \underline{v_1' = -8 + 3 = -5} \\ \underline{v_2' = 1 + 5 = 6} \\ \underline{v_3' = 5} \end{array}$$

$$(7) (iv) A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix} \quad (B)$$

$$A^{-1} =$$

$$[A|I] = \left[\begin{array}{ccc|ccc} 3 & 4 & -1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 2 & 5 & -4 & 0 & 0 & 1 \end{array} \right]$$

$$R_1 \leftrightarrow R_2 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 3 & 4 & -1 & 1 & 0 & 0 \\ 2 & 5 & -4 & 0 & 0 & 1 \end{array} \right]$$

$$\begin{aligned} R_2' &= -3R_1 + R_2 \\ R_3' &= -2R_1 + R_3 \end{aligned} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 4 & -10 & 1 & -3 & 0 \\ 0 & 5 & -10 & 0 & -2 & 1 \end{array} \right]$$

$$R_3' = -R_2 + R_3 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 4 & -10 & 1 & -3 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \end{array} \right]$$

$$R_2 \leftrightarrow R_3 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 4 & -10 & 1 & -3 & 0 \end{array} \right]$$

$$R_2' = -4R_2 + R_3 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & -10 & 5 & -7 & -4 \end{array} \right]$$

$$R_3' = -\frac{1}{10}R_3 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{array} \right]$$

$$\underline{v_1' = -3v_2 + v_1} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/2 & -1/10 & -6/5 \\ 0 & 1 & 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & -1/2 & 7/10 & 2/5 \end{array} \right]$$

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